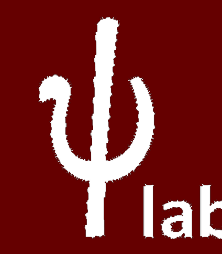


# ECGluFormer: glucose prediction from ECG via multi-loss, transformer-based aggregation



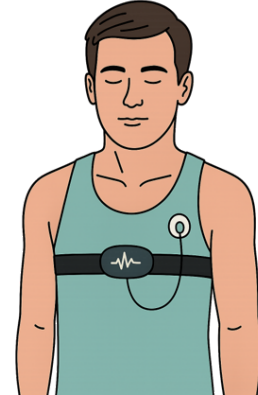
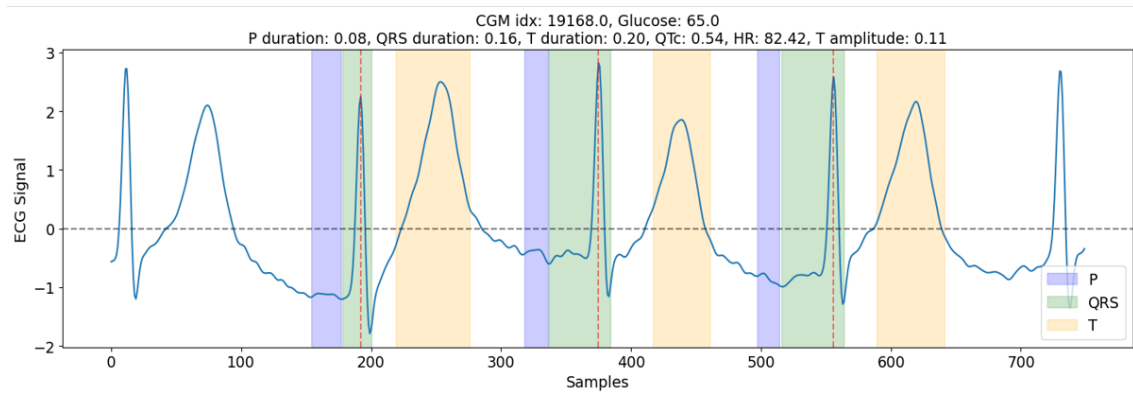
Mu-Ruei Tseng    Ricardo Gutierrez-Osuna

## Introduction

- Diabetes affects 600M+ people worldwide
- CGMs are the main technology to monitor glucose, but they are **invasive, costly**, and can cause **skin irritation**

⇒ Critical Need: Non-invasive continuous glucose monitoring with accurate hypoglycemia detection

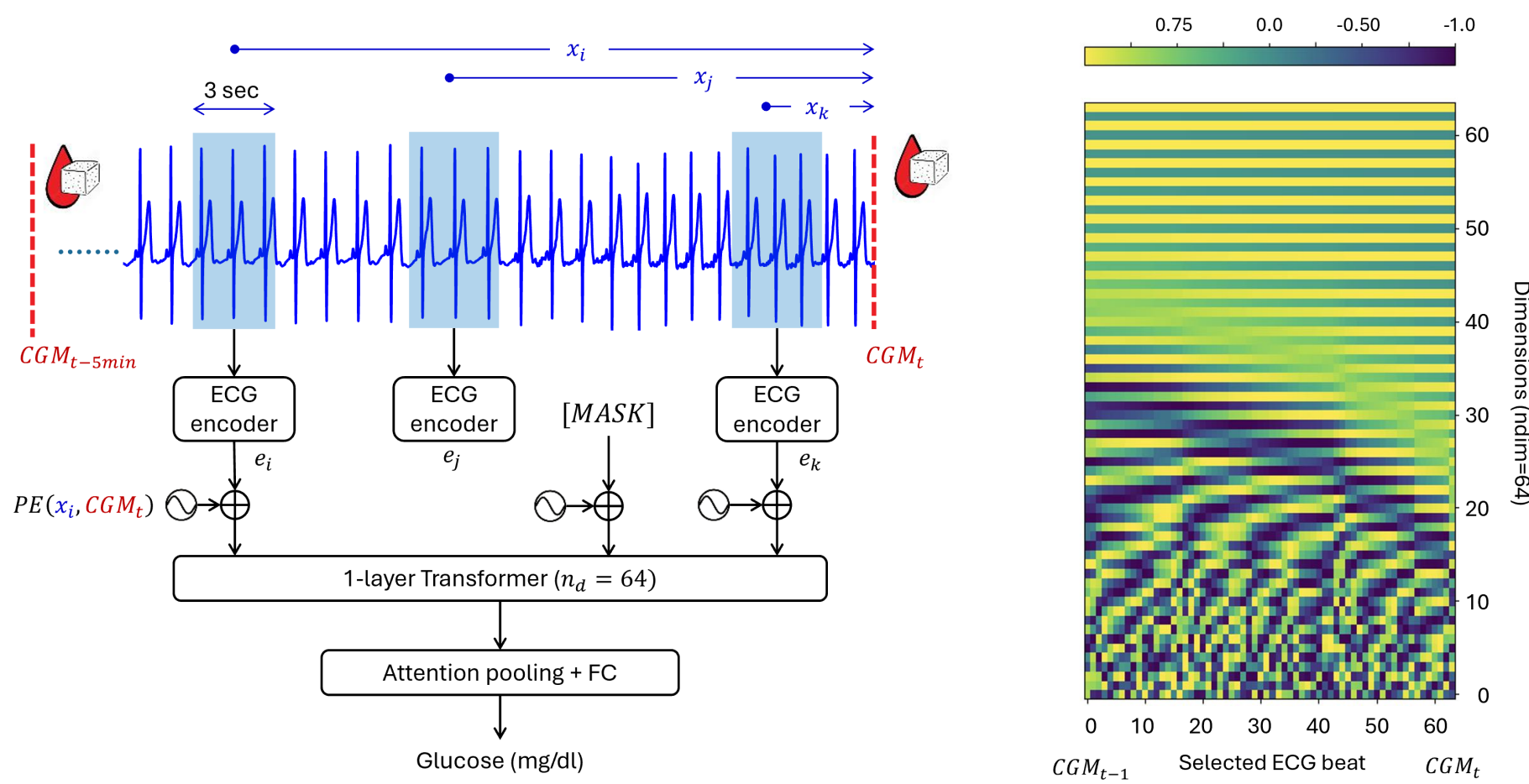
⇒ Electrocardiograms (ECG) are robust to motion artifacts and capture changes in beat morphology associated with glycemic changes



Problem:

- Data imbalance:** < 4% CGM readings are < 70mg/dL, leading MSE loss to under-report these critical events
- Intermittent data:** Connectivity issues can cause data loss in free-living conditions
- Temporal misalignment:** ECG beats and CGM have different sampling rates and are not synchronized

## Proposed Method



### Stage 1: Beat-level encoder with multi-objective loss

We extract 3-sec ECG beats (R-peak centered) using NeuroKit2, label it with nearest future CGM value

We train an InceptionTime-based encoder [1] using a multi-objective loss that addresses data imbalance and under-reporting of critical hypoglycemic events:

$$\mathcal{L}(\hat{G}_{\text{batch}}, G_{\text{batch}}) = \underbrace{\text{MSE}(\mathcal{S}_{\hat{G}_{\text{batch}}}, \tilde{G}_{\text{pseudo}})}_{\text{distribution loss}} + \lambda \cdot \underbrace{\text{NMSE}(\hat{G}_{\text{batch}}, G_{\text{batch}})}_{\text{sample-wise loss}} \quad (1)$$

$$\text{where } \text{NMSE}(\hat{G}, G) = \frac{1}{n} \sum \frac{\|\hat{g} - g\|^2}{\|g\|^2} \times \frac{1}{f_g}$$

- Distribution loss:** aligns prediction distribution with ground truth via pseudo-labels  $\tilde{G}_{\text{pseudo}}$
- NMSE with inverse frequency weighting ( $1/f_g$ ):** emphasizes errors in hypoglycemic range

### Stage 2: CGM-level aggregation via transformer

We aggregate beat-level embeddings over 5-min windows using a Transformer with:

- Time-based positional encoding:**  $\text{pos}_i = t(\text{CGM}_t) - t(x_i)$  with sinusoidal encoding (actual timestamps, not sequential indices)
- Sampling:** 64 beats/window (34% data loss), sorted by time
- BERT masking:** 10% tokens masked for robustness
- Attention pooling:**  $\alpha_t = \frac{\exp(w^\top h_t)}{\sum_{t'} \exp(w^\top h_{t'})}$ ,  $z = \sum_t \alpha_t h_t$  focuses on salient beats
- End-to-end training:** Encoder refined with Transformer

In this fashion, the model can capture temporal dynamics despite 30%+ data loss

## Dataset

PhysioCGM: 10 participants with type-1 diabetes, up to 17 days each

- Sensors:** Zephyr Bioharness (ECG) + Dexcom G6 (CGM), synchronized
- Split:** 70/15/15 train/val/test with stratified sampling to preserve hypo/normo ratio
- Model:** Subject-specific (one model per participant)

## Results

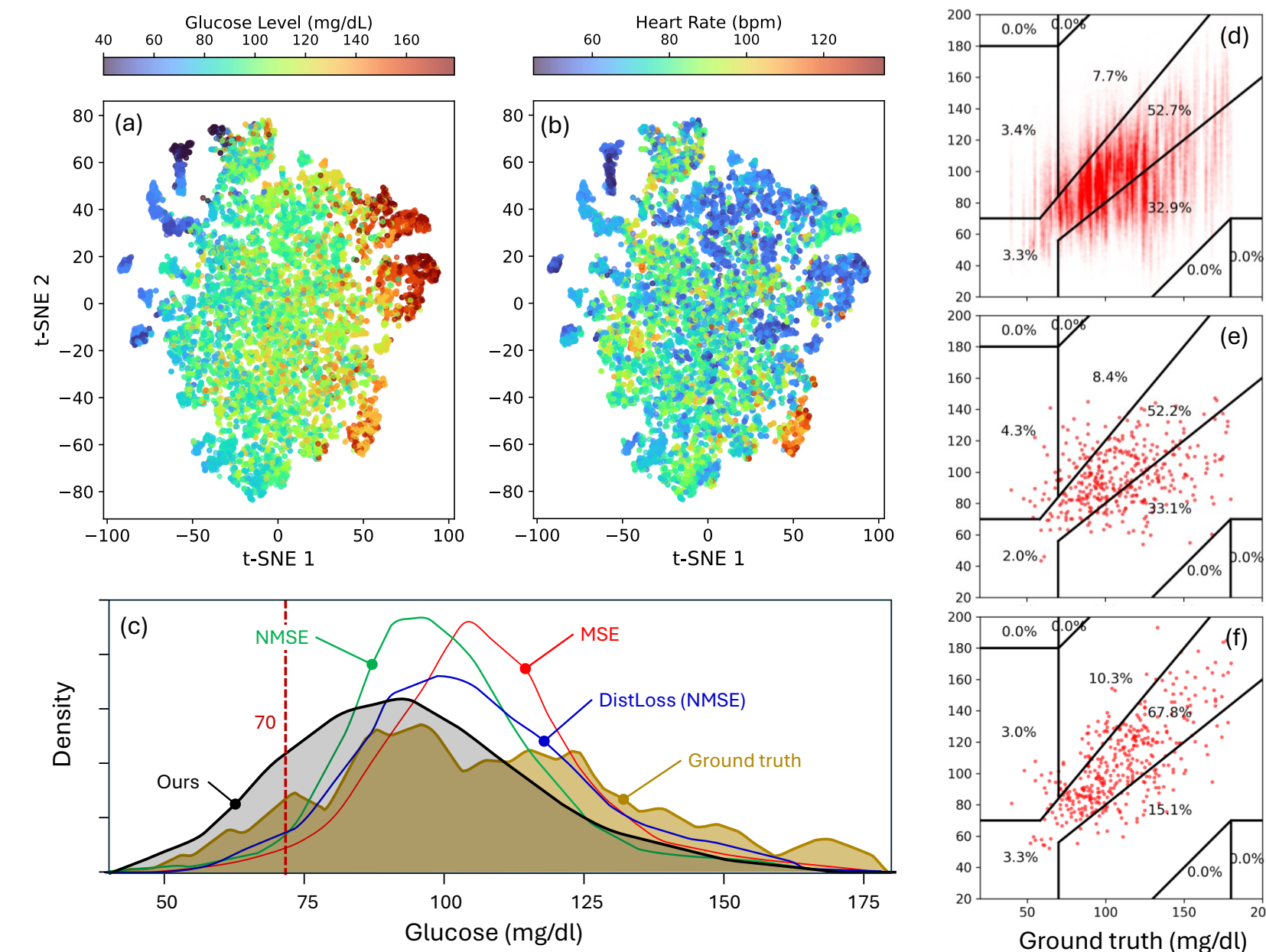


Figure 1. **Model Analysis.** (a-b) t-SNE visualization of ECG encoder embeddings colored by glucose level and heart rate. (c) Ground truth vs. predicted glucose distributions for different loss functions. (d-f) Clarke Error Grid comparing beat-level predictions, baseline B2 [2], and ECGluFormer.

### Clarke Error Grid Performance

98.99%  
Zone A+B  
Clinically Safe

1.00%  
Zone D  
Failure to Detect

### Model Comparison (Average Performance)

| Model         | Zone A+B↑ | Zone D↓ | Sens.↑ | Spec.↑ | RMSE↓ |
|---------------|-----------|---------|--------|--------|-------|
| ECGluFormer   | 98.99%    | 1.00%   | 0.86   | 0.85   | 20.83 |
| LSTM          | 98.47%    | 1.52%   | 0.82   | 0.78   | 23.14 |
| Histogram+FC  | 98.20%    | 1.79%   | 0.84   | 0.80   | 21.32 |
| Porumb et al. | 96.63%    | 3.34%   | 0.72   | 0.68   | 28.79 |
| Feature-based | 95.61%    | 4.39%   | 0.75   | 0.56   | 24.40 |

### Loss Function Ablation Study

| Loss Function     | Zone D↓ | Sens.↑ | Spec.↑ | RMSE↓ |
|-------------------|---------|--------|--------|-------|
| MSE               | 3.04%   | 0.76   | 0.72   | 19.82 |
| NMSE              | 2.45%   | 0.79   | 0.76   | 22.21 |
| DistLoss          | 2.41%   | 0.80   | 0.77   | 21.61 |
| Ours (Multi-Loss) | 1.59%   | 0.80   | 0.77   | 27.17 |

## Conclusion

- First transformer model for ECG-based glucose prediction with time-aware beat aggregation
- Robust to 30%+ data loss in free-living conditions
- 48% reduction in Zone D errors vs. conventional MSE loss
- Lightweight: 410K params, 8.2ms inference (GPU)

## References

- Hassan Ismail Fawaz, Benjamin Lucas, Germain Forestier, Charlotte Pelletier, Daniel F. Schmidt, Jonathan Weber, Geoffrey I. Webb, Lhassane Idoumghar, Pierre-Alain Muller, and François Petitjean. Inceptiontime: Finding alexnet for time series classification. *Data Min Knowl Discov*, 34(6):1936–1962, 2020.
- Mihaela Porumb, Saverio Stranges, Antonio Pescapè, and Leandro Pecchia. Precision medicine and artificial intelligence: a pilot study on deep learning for hypoglycemic events detection based on ecg. *Sci Rep*, 10(1):170, 2020.